

Week 14 – Full Data Preprocessing Pipeline & Verification

Objective

The objective of Week 14 was to design, implement, and verify a complete data preprocessing pipeline supporting multimodal deepfake detection. This pipeline integrates video frames, audio waveforms, and audio features (MFCCs) across multiple heterogeneous datasets, ensuring reproducibility, robustness, and storage efficiency.

Datasets integrated at this stage:

- DeeperForensics
- FakeAVCeleb v1.2

Preprocessing Pipeline Overview

The preprocessing pipeline was designed as a dataset-agnostic, CSV-driven system, enabling consistent processing across datasets with different directory structures and modalities.

Pipeline Stages:

1. Dataset manifest generation
2. Train / validation / test split creation
3. Video frame extraction
4. Audio stream extraction
5. MFCC feature generation
6. Integrity verification

Each stage is implemented as an independent script to ensure modularity and auditability.

Video Frame Extraction

Method:

- A fixed number of 5 frames per video are extracted.
- Frames are uniformly sampled across the full video duration.
- Extraction is resume-safe (already-processed videos are skipped).

Rationale:

- Uniform sampling avoids temporal bias.
- Limiting to 5 frames significantly reduces storage requirements while preserving visual diversity.
- This strategy aligns with common deepfake detection baselines in the literature.

Output Structure:

data/interim/frames/<Dataset>/<split>/<video_id>/

frame_000001.jpg

...

frame_000005.jpg

Audio Extraction

Tooling:

- FFmpeg (Essentials Build) for audio decoding
- ffprobe for audio stream detection

Dataset-Specific Behavior:

- DeeperForensics: No audio streams detected → audio extraction intentionally skipped.
- FakeAVCeleb v1.2: Audio successfully extracted for all applicable videos.

This conditional behaviour demonstrates pipeline robustness to missing modalities.

Output Structure:

data/interim/audio/<Dataset>/<split>/<video_id>.wav

MFCC Feature Generation

Method:

- MFCCs computed from WAV files using:
 - 40 MFCC coefficients.
 - Standard windowing and hop length.
- Output saved as NumPy arrays.

Output Structure:

data/processed/audio_features/<Dataset>/<split>/<video_id>.npy

MFCC matrices have shape:

(40, T) where T varies with audio duration, confirming real temporal information preservation.

Dataset Integration

DeeperForensics:

- Modalities used: Video only.
- Audio intentionally excluded due to dataset characteristics.

- Suitable for visual-only deepfake detection

FakeAVCeleb v1.2:

- Modalities used: Video + Audio
- Supports:
 - FakeVideo–FakeAudio
 - FakeVideo–RealAudio
 - RealVideo–FakeAudio
 - RealVideo–RealAudio
- Enables future audio-visual fusion experiments.

Preprocessing Integrity Verification

A dedicated integrity-checking script was executed to validate preprocessing outputs.

DeeperForensics (Validation Split):

- 738 video folders verified.
- Each folder contains exactly 5 frames.
- 0 audio streams detected (confirmed via ffprobe).
- Audio + MFCC intentionally skipped.

FakeAVCeleb v1.2 (Validation Split):

- Randomly sampled items verified:
 - WAV file present
 - MFCC feature present
 - MFCC shape consistent ($40 \times T$)
- 10/10 sampled items passed.

All preprocessing outputs were verified as correct, complete, and consistent with design specifications.

Methodology Summary

Design Principles:

- Reproducibility: CSV-driven pipeline, deterministic sampling
- Modularity: Independent scripts per preprocessing stage
- Scalability: Dataset-agnostic logic
- Efficiency: Lightweight frame sampling and feature storage

Data Flow:

Raw Videos => Dataset Manifests (CSV) => [Frame Extraction -> Interim Frames] => [Audio Extraction -> WAV Files] => [MFCC Extraction -> Feature Arrays] => Integrity Verification

Model Design

Q1. Why extract only 5 frames per video?

The goal is to balance representational diversity with storage efficiency. Uniform sampling across the full video duration captures different facial poses and lighting conditions while avoiding the exponential storage cost of full-frame extraction. This approach is widely used in baseline deepfake detection systems.

Q2. Why use MFCCs instead of raw audio?

MFCCs provide a compact, perceptually meaningful representation of speech. They reduce input dimensionality while preserving frequency patterns relevant to synthetic speech artefacts, making them suitable for classical and deep learning audio classifiers.

Q3. Why skip audio for DeeperForensics?

DeeperForensics videos contain no audio streams. The pipeline explicitly detects this using ffprobe and skips audio processing intentionally, demonstrating robustness rather than failure.

Q4. Why separate interim and processed data?

This separation enforces a clean data lifecycle:

- Interim: decoded but untransformed artefacts (frames, WAVs)
- Processed: model-ready features (MFCCs)

This improves traceability, debugging, and reproducibility.

Q5. How does this pipeline support future multimodal models?

The pipeline produces synchronised visual and audio features indexed by video ID. This enables late fusion, early fusion, or cross-modal attention architectures without reprocessing raw data.

Week 14 Completion Statement

All Week 14 objectives have been fully completed:

- Full preprocessing pipeline implemented.
- Multi-dataset integration achieved.
- Audio and visual modalities processed correctly.
- Integrity verified through automated checks.

The system is now ready for advanced multimodal model development and experimentation.